

Ambient WiFi as a Radar Replacement for Automotive SLAM: A Physics-Based Feasibility Study

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Abstract—Automotive radar is a costly, dedicated sensor, while WiFi transceivers are ubiquitous and, under the integrated sensing and communication (ISAC) paradigm, increasingly dual-purpose. We ask whether *ambient* sub-7 GHz WiFi—received on a single moving-vehicle antenna and processed with commodity channel-state information (CSI)—can serve in place of radar as the perception front-end of a SLAM pipeline. Using physics-based ray tracing (Sionna RT), we build an end-to-end pipeline: ray-traced multipath channel $\rightarrow$ super-resolution (MUSIC) delay/angle estimation $\rightarrow$ bistatic reflector triangulation $\rightarrow$ Rao-Blackwellized particle-filter SLAM, and evaluate it with trajectory (ATE/RPE) and map (Chamfer, occupancy IoU, directional accuracy/completeness) metrics. We find that ambient WiFi *localizes* the vehicle to centimetre level with a clean operating envelope that follows the $\Delta R = c/2B$ resolution law, and that a joint two-dimensional (delay-angle) MUSIC front-end lifts *realistic* commodity-CSI localization to oracle-sensing quality in sparse-multipath scenes. In contrast, *mapping* separates into two tiers: with clean single-bounce (oracle) sensing the back-end reconstructs reflecting surfaces to 25–30 cm, whereas realistic sensing is bounded to 4–5 m. Counter-intuitively, this floor is removed by *neither* a $44\times$ bandwidth increase (60 GHz) *nor* a $4\times$ larger array: the limit is not resolution but *path discrimination*—telling a genuine reflection from a line-of-sight or floor path—which commodity CSI does not provide. These results delineate precisely what passive WiFi can and cannot do for automotive SLAM and identify bounce/path discrimination as the key open problem for WiFi mapping.

Index Terms—WiFi sensing, passive WiFi radar, channel state information, SLAM, integrated sensing and communication, automotive perception, multipath-based SLAM, Sionna RT.

I. INTRODUCTION

AUTOMOTIVE perception relies heavily on radar for its robustness to weather and lighting, but radar is a dedicated, cost- and power-bearing sensor. In parallel, WiFi radios are ubiquitous—in vehicles, roadside units and the built environment—and the emerging ISAC paradigm and the IEEE 802.11bf sensing amendment are turning communication hardware into opportunistic sensors [1], [2], [3]. This raises a concrete question for autonomous systems: *can ambient WiFi, received on a moving vehicle, replace radar as the perception input to SLAM?*

The gap. A two-round, adversarially fact-checked survey (summarized in Sec. II) exposes a clean and defensible gap.

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Code, configurations and results are openly available and citable: GitHub <https://github.com/mulhamfetna/wifi-radar-slam>; Zenodo concept DOI 10.5281/zenodo.21247288.

Passive WiFi radar and CSI-based sensing are mature but almost exclusively *indoor, static and infrastructure-fixed* [4], [5], [6], [7]; the one demonstrated *moving-platform* passive radar uses a 5G downlink—not WiFi—as its illuminator [8]; and joint communication and SLAM is explicitly “in its infancy” [9]. No prior work occupies the intersection of an *on-vehicle, mobile, outdoor, WiFi-band* sensing system feeding a SLAM pipeline.

The physical fork. Range resolution is bandwidth-limited, $\Delta R = c/2B$: commodity sub-7 GHz WiFi resolves only 3.75 m at 40 MHz and 0.94 m at 160 MHz, whereas radar-grade resolution (~ 8.5 cm) requires 60 GHz mmWave (IEEE 802.11ad DMG) [10], [1]. A WiFi-for-SLAM design must therefore either (i) stay sub-7 GHz and extract coarse geometry from multipath, or (ii) move to 60 GHz for fine range. *This paper takes the sub-7 GHz path and characterizes it honestly and end-to-end*, isolating what is achievable now from what needs the mmWave extension.

Contributions.

- An open, reproducible, physics-based (Sionna RT ray-traced) end-to-end pipeline for on-vehicle ambient-WiFi SLAM, spanning channel simulation, super-resolution sensing, bistatic mapping and particle-filter SLAM.
- A demonstration that ambient sub-7 GHz WiFi *localizes* a moving vehicle to centimetre level, with an operating envelope (bandwidth, SNR, access points, speed) that quantitatively follows the $\Delta R = c/2B$ law.
- A *joint two-dimensional (delay-angle) MUSIC* front-end whose intrinsic association lifts realistic commodity-CSI localization to oracle-sensing quality in sparse multipath.
- A two-tier characterization of *mapping*: an oracle upper bound (25–30 cm) and a realistic bound (4–5 m), with the residual gap shown—by a 60 GHz and a 16-antenna test—to be a *path-discrimination* limit of commodity CSI rather than a bandwidth or aperture (resolution) limit, identifying the key open problem for WiFi mapping.
- **WiFiSLAM-Sim**, the first (ray-traced) outdoor/vehicular WiFi-CSI dataset with per-path labels, and a learned discriminator that predicts mapping-useful reflections from estimable features ($F1 \approx 0.9$ at realistic noise)—a first concrete step on the path-discrimination problem.

II. RELATED WORK

A. Passive WiFi Radar

Ambient WiFi access points act as illuminators of opportunity for bistatic passive radar. Chetty *et al.* first demonstrated

through-wall detection of moving personnel, extracting both range and Doppler [4]. Because WiFi bursts have short integration time and low bandwidth, standard Fourier processing is inadequate, motivating super-resolution (ESPRIT/MUSIC) methods [5]; passive radar has been shown against unmodified stand-alone access points [6], and phase-robust CSI-power processing achieves sub-metre indoor tracking [11]. All of this work is indoor and static—the point of departure for a mobile outdoor system.

B. WiFi CSI Sensing and 3D Reconstruction

WiFi-band signals reflect off bodies and penetrate walls; learning-based methods reconstruct dense structure—2D/3D human pose [0], [0], [7], [0], depth [0], and even latent-diffusion imagery [0]. These systems prove dense WiFi “imaging” is possible, but every one is a few-metre *fixed indoor* zone with poor cross-environment generalization [7]—which both motivates and bounds the present work. We are careful to distinguish purpose-built WiFi-band FMCW radios [0] from decoded commodity 802.11 CSI [0].

C. RF- and Multipath-based SLAM

The SLAM back-end we require already exists. Channel-SLAM treats each multipath component as a line-of-sight signal from a *virtual transmitter*, jointly estimating landmark and receiver states with no prior map [0], [0]; modern multipath-based SLAM extends this with cooperation, map fusion and belief propagation, including real mmWave-MIMO measurements [0], [0]. We reuse this virtual-transmitter / bistatic-reflector landmark model and test its robustness to a fast-moving receiver.

D. ISAC, IEEE 802.11bf and mmWave WiFi

ISAC formalizes the shared-hardware sensing/communication tradeoff [2], [0], [3], and IEEE 802.11bf standardizes the sensing measurement plumbing while leaving algorithms to vendors [1]. Crucially for our design fork, the 802.11ad 60 GHz waveform has been reused as a radar with <0.1 m range resolution and V2I imaging, albeit in theory/simulation [10], [0]. This frames our sub-7 GHz-now, 60 GHz-next roadmap.

III. SYSTEM MODEL AND METHOD

A. Scenario and Bistatic Geometry

A vehicle carrying a single receive antenna array drives through an outdoor scene illuminated by N_{AP} ambient WiFi access points at known positions. Each specular multipath component follows an access point $\rightarrow$ reflector $\rightarrow$ vehicle path, so the measured path length $\rho = \|\mathbf{p}_{\text{AP}} - \mathbf{r}\| + \|\mathbf{r} - \mathbf{p}_{\text{veh}}\|$ places the reflector $\mathbf{r}$ on an ellipse with foci at the access point and the vehicle. Given the arrival azimuth ϕ (the bearing from the vehicle to $\mathbf{r}$), the reflector range s along $\hat{\mathbf{u}}(\phi)$ solves

$$s = \frac{\rho^2 - d_{\text{AP}}^2}{2(\rho - \mathbf{d}_{\text{AP}} \cdot \hat{\mathbf{u}}(\phi))}, \quad \mathbf{d}_{\text{AP}} = \mathbf{p}_{\text{AP}} - \mathbf{p}_{\text{veh}}, \quad (1)$$

with $d_{\text{AP}} = \|\mathbf{d}_{\text{AP}}\|$; the direct path ($s \leq 0$) and physically implausible solves are rejected.

B. Ray-Traced Channel Model

Channels are generated with Sionna RT 2.0 (Mitsuba/Dr.Jit), which ray-traces the scene (buildings, vehicles) to produce per-path delays, angles, interaction types and the frequency-domain channel (CFR) across N_{sc} OFDM subcarriers and N_{rx} antennas, per vehicle pose along the trajectory. Additive white Gaussian noise is applied at a configured SNR.

C. Sensing Front-End

We compare two sensing tiers. The *oracle* front-end reads Sionna’s true single-scatter path parameters (delay, azimuth), isolating the mapping/SLAM geometry from estimation error. The *realistic* front-end estimates parameters from the noisy CSI with MUSIC. For delays we use the antennas as snapshots over a physically-bounded delay grid; for azimuth we use the subcarriers as snapshots. A key refinement is *joint 2-D (delay-angle) MUSIC* with 2-D spatial smoothing (SVD subspace; delay grid bounded to the unambiguous $1/\Delta f$ range), which recovers each path’s delay *and* angle together, so their association is intrinsic rather than paired by sorted index. Array-relative electrical angles are mapped to world azimuth by $\sin \beta = -\sin \theta$ for the vehicle’s lateral ULA.

D. SLAM Back-End and Metrics

A Rao–Blackwellized particle filter propagates the vehicle pose from odometry and weights particles by the bistatic consistency (1) of each detection; triangulated reflectors are accumulated and consensus-filtered into a landmark map. We report trajectory error (ATE, RPE) and, for the map, symmetric Chamfer distance, occupancy IoU, and *directional* accuracy (estimate $\rightarrow$ ground truth) and completeness (ground truth $\rightarrow$ estimate), since a passive-WiFi map illuminates only a subset of surfaces. Ground truth for the map is the facade footprint of each scene object.

IV. EXPERIMENTAL SETUP

Two scene families are used. A *street canyon* (Sionna’s `simple_street_canyon_with_cars`) provides a realistic, dense-multipath outdoor environment; a *controlled reflective* scene (a scaled metal wall) provides a geometry in which clean single-bounce returns are guaranteed, isolating sensing from illumination effects. Unless stated otherwise: carrier 5.2 GHz, 40 MHz bandwidth, $N_{\text{sc}} = 64$, $N_{\text{rx}} = 4$ (half-wavelength ULA), 3 access points, 20 dB SNR, 5 m/s straight trajectory. Simulations run CPU-only (Sionna RT LLVM back-end) on a 32-thread workstation; each Phase-A run completes in tens of seconds. All numbers below are reproduced from the released result artifacts (`docs/results-v1.md`).

V. RESULTS

A. Localization Operating Envelope

Ambient WiFi localizes the vehicle to centimetre level. Table I sweeps the four principal parameters. Bandwidth shows the expected monotonic $\sim 20\times$ ATE improvement from 20 MHz to 160 MHz, tracking the $\Delta R = c/2B$ law end-to-end; SNR degrades gracefully to sub-decimetre by 30 dB; more access points and higher speeds are generally beneficial.

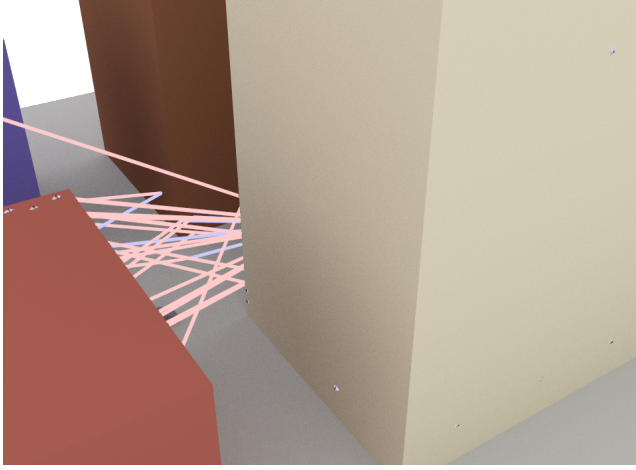


Fig. 1. Ray-traced street-canyon scene with the ambient-WiFi multipath rays from an access point to the vehicle antenna (Sionna RT render).

TABLE I
LOCALIZATION OPERATING ENVELOPE: ABSOLUTE TRAJECTORY ERROR (ATE, M), ONE PARAMETER SWEPT AT A TIME ABOUT THE NOMINAL POINT.

Bandwidth		SNR	
20 MHz	0.78	0 dB	2.02
40 MHz	0.093	10 dB	1.86
80 MHz	0.058	20 dB	1.01
160 MHz	0.038	30 dB	0.076
# Access points		Speed	
1	0.084	1 m/s	2.13
2	1.97	5 m/s	0.032
3	0.075	10 m/s	0.047
4	0.041	15 m/s	0.010

B. Realistic Sensing: Joint Delay–Angle MUSIC

Separate 1-D delay and 1-D angle estimates paired by sorted index mis-associate in multipath and scatter phantom reflectors. Joint 2-D MUSIC removes this by estimating each path’s delay and angle together. In the controlled scene at 160 MHz, realistic ATE improves from 0.73 m (sorted) to 0.027 m (joint), matching the 0.045 m oracle bound—i.e. *realistic commodity-CSI localization becomes oracle-quality* in sparse multipath. The benefit is scene-dependent: in the dense street canyon (~ 19 paths, 4 antennas) the paths cannot be resolved and joint estimation does not help, so it is retained as an opt-in front-end.

C. Mapping: Oracle vs. Realistic

Table II contrasts the two sensing tiers. With oracle sensing the map reconstructs reflecting surfaces to 25–30 cm (Fig. 2, left); the same geometry under realistic MUSIC sensing is bounded to 4–5 m. Critically, this residual persists even in the front/back-unambiguous controlled scene and even under joint estimation; as Sec. VII shows, it is removed by neither wider bandwidth nor larger aperture. The bound is therefore a *path-discrimination* limit—genuine facade reflections cannot be separated from the consistent LOS/floor paths commodity

TABLE II
MAPPING ACCURACY: ORACLE VS. REALISTIC SENSING.

Scene / sensing	accuracy (m)	Chamfer (m)	IoU
Controlled wall, oracle	0.25	0.51	0.79
Street canyon, oracle	0.30	12.3 [†]	0.08
Controlled wall, realistic (joint)	4.8	4.1	0.0

[†] symmetric Chamfer inflated by partial coverage; a single WiFi pass illuminates only street-facing facades.

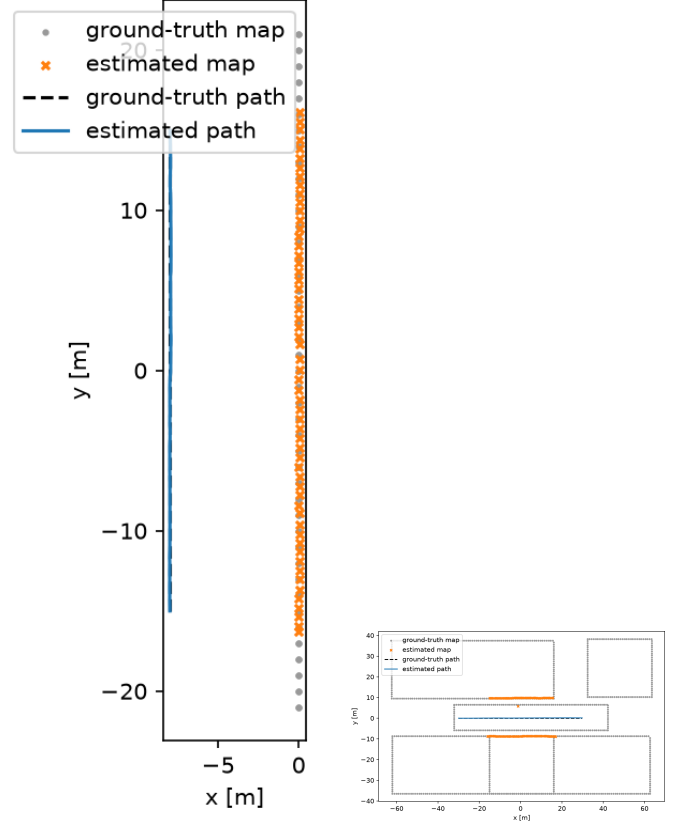


Fig. 2. Oracle maps. Left: controlled wall—estimated reflectors (orange) trace the wall to ~ 0.25 m. Right: reflective street canyon—reflectors trace the two illuminated street-facing facades; distant/outer surfaces are uncovered (coverage, not accuracy, limited).

CSI returns without an interaction label—not a resolution problem.

D. Real Commodity-CSI Proof-of-Concept

To verify that the sensing front-end is not simulation-specific, we ran the same MUSIC delay/AoA estimator on measured hardware CSI through a CSIKit-backed adapter. On Intel 5300 captures (30 subcarriers, 3 antennas) and Broadcom nexmon 802.11ac captures (80 MHz), the front-end runs end-to-end and returns plausible indoor multipath delay/angle estimates. This is a front-end validation only—the captures are indoor and static with no ground-truth trajectory—but it confirms the pipeline consumes real commodity CSI. A full outdoor, vehicle-mounted validation is blocked by the absence of any public outdoor/vehicular WiFi-CSI dataset (the very gap this work targets) and is the key next step.

TABLE III
NEITHER BANDWIDTH NOR APERTURE MOVES REALISTIC MAP
ACCURACY.

Configuration (controlled wall, joint)	accuracy (m)	ATE (m)
sub-7 GHz, 40 MHz, 4-antenna	4.8	0.027
60 GHz, 1.76 GHz, 4-antenna	4.8	0.030
60 GHz, 1.76 GHz, 16-antenna	4.75	0.034

VI. DISCUSSION

The results delineate a clear boundary. *Localization* from ambient sub-7 GHz WiFi is practical today: centimetre-level and, with joint estimation, oracle-quality from realistic CSI in favourable geometry. *Mapping* is fundamentally harder, but—as Sec. VII shows—not for the reason one would expect: the realistic floor is removed by neither wider bandwidth nor larger aperture, so it is not a resolution limit but a *path-discrimination* limit. Genuine facade reflections cannot be separated from the consistent line-of-sight and floor paths that commodity CSI returns without an interaction label. Limitations of this study are its simulation-only scope and the single-array receiver.

VII. DOES MORE BANDWIDTH OR APERTURE FIX MAPPING? A 60 GHz TEST

The intuitive remedy for the mapping floor is finer resolution: a 60 GHz (IEEE 802.11ad DMG) front-end offers ~ 1.76 GHz of bandwidth ($\Delta R \approx 8.5$ cm, a $44\times$ improvement), and a larger array sharpens angle. We tested both directly by re-running the controlled-wall pipeline at 60 GHz and with a 16-element array (Table III).

Strikingly, neither change moves the map accuracy (Fig. 3). The reason is visible in the reconstructed map: the estimated reflectors form a *consistent phantom arc along the vehicle trajectory*, not the wall—these are the line-of-sight and floor-bounce paths, which MUSIC returns among the strongest components at every pose and which, being geometrically consistent, survive consensus filtering. Bandwidth and aperture sharpen *resolution*, but the mapping floor is *path discrimination*: distinguishing a genuine facade reflection from an LOS/floor/multi-bounce path. The oracle succeeds only because it can filter by the ray tracer’s interaction type; commodity CSI carries no such label.

Consequence. The realistic-mapping limit is neither the $\Delta R = c/2B$ resolution ceiling nor array aperture—so 60 GHz alone does not enable passive-WiFi mapping. It is the absence of bounce/path discrimination in commodity CSI, which we identify as the key open problem. Localization, which does not require this discrimination, is unaffected and remains centimetre-level at both bands.

As a first step toward discrimination, we gate on the bistatic *excess* path length (LOS and floor paths barely detour, so their excess is near zero). This halves the phantom error (map accuracy 4.8 m $\rightarrow$ 2.0 m) but collapses coverage, because the delay bias makes genuine reflections low-excess too and they are gated out alongside the phantoms.

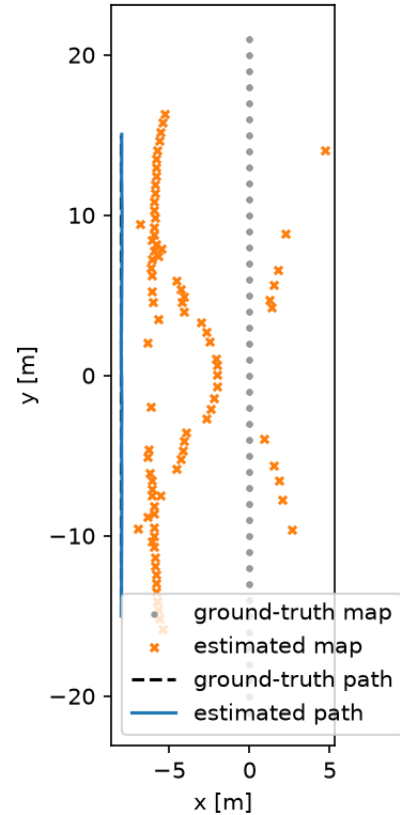


Fig. 3. Realistic 60 GHz map (joint MUSIC). Despite 8.5 cm range resolution, the estimated reflectors (orange) form a consistent phantom arc along the vehicle trajectory (LOS/floor paths), not the wall (grey, at $x=0$).

A. Learned Path Discrimination

Is the discrimination *learnable*? We release **WiFiSLAM-Sim**, a ray-traced outdoor/vehicular WiFi-CSI dataset that—uniquely, being simulated—carries a per-path oracle label (bounce count, interaction type, floor flag) usable as a supervised target. A random forest predicts the binary label “*mapping-useful single-scatter facade reflection*” from per-path features a receiver can estimate (range, bistatic excess, azimuth, elevation, azimuth-deviation-from-AP)—not the interaction type. On exact features the classes are perfectly separable ($F1 = 1.00$, excess and azimuth-deviation dominating); under realistic MUSIC-level feature error it degrades gracefully— $F1 = 0.94/0.90/0.86$ at range/azimuth noise of $(1\text{ m}, 3^\circ)/(2\text{ m}, 6^\circ)/(4\text{ m}, 10^\circ)$ —well above the fixed excess gate. Path discrimination is thus *learnable and robust*; its ceiling is set by feature-estimation accuracy (chiefly the delay-derived excess), closing the loop with the delay-bias finding.

VIII. FUTURE WORK

Three directions follow directly. (i) *Feature estimation for path discrimination*: our learned discriminator is accurate given estimable features, so the remaining problem is estimating them (chiefly the delay-derived excess) robustly from noisy CSI—e.g. end-to-end learning directly on CSI rather than on MUSIC outputs. (ii) *A real outdoor, vehicle-mounted WiFi-CSI capture*: WiFiSLAM-Sim supplies a simulated dataset with

labels, but real-hardware outdoor CSI still does not exist and is the prerequisite for full-system (not just front-end) validation. (iii) Tighter integration with a virtual-transmitter multipath-SLAM back-end [0], [0] and Doppler exploitation for moving clutter.

IX. CONCLUSION

We presented an open, physics-based feasibility study of ambient sub-7 GHz WiFi as a radar replacement for automotive SLAM. Ambient WiFi localizes a moving vehicle to centimetre level—and to oracle-sensing quality from realistic CSI via joint delay–angle MUSIC in sparse multipath—while mapping separates into an oracle upper bound (25–30 cm) and a realistic bound (4–5 m). We further showed that this mapping floor is removed by neither a 60 GHz wide-band front-end nor a larger array—it is a *path-discrimination* limit of commodity CSI, not a resolution one. The study thus maps the boundary of the achievable for passive WiFi perception and identifies bounce/path discrimination as the central open problem for WiFi-based mapping.

REPRODUCIBILITY

All code, configurations, scenes and result artifacts are openly available: GitHub <https://github.com/mulhamfetna/wifi-radar-slam>; archived on Zenodo (concept DOI 10.5281/zenodo.21247288).

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